

Modulation-speed workload counterexamples for Cox/D/1 queues

Tynan Daly

10 September 2026

Status. Mathematical derivation with exact-arithmetic certificates and numerical cross-checks, not independent peer review or formal verification. Novelty is not established. The target is the universal workload inequality in Problem 1 of Leskelä (2022), including its restriction to finite-state Markov intensity processes. Established positive theorems for special ordered Markov environments are not contradicted.

1 Main statement

There is a 64-state continuous-time Markov intensity process with strictly positive intensity levels and strictly positive transition rates between every pair of distinct states such that increasing the modulation speed from 2π to 3π strictly increases stationary mean workload in the unit-service, unit-speed single-server queue. An explicit bound is

$$w(3\pi) - w(2\pi) > \frac{1587}{25\,580\,800\,000\,000} > 0.$$

The environment is stationary, irreducible and mixing. The statement does not rely on a periodic sample path, non-Markov hidden phases, zero intensities or zero transition rates. A separate 24-state example below has entirely rational parameters, though a smaller certified effect; its transition graph is a directed cycle. Neither state count is claimed globally minimal.

2 The workload inequality used in both certificates

Let X be stationary with $\mathbb{E}X(0) = 1$ and $0 \leq X \leq M$. Conditional on X , let arrivals be Poisson of intensity $\epsilon X(ct)$. Every service requirement is exactly 1, the server speed is 1, and $\epsilon M < 1$. Define

$$K(c) = \int_0^1 (1-u) \mathbb{E}[X(0)X(cu)] du.$$

The stationary mean workload satisfies

$$\frac{\epsilon}{2} + \epsilon^2 K(c) \leq w(c) \leq \frac{\epsilon}{2} + \frac{\epsilon^2 K(c)}{1 - \epsilon M}. \tag{1}$$

Construction and integrability

Use a two-sided stationary environment and an independent homogeneous Poisson proposal process of rate ϵM with independent uniform marks. Accept a point at t when its mark is at most $X(ct)/M$. Start empty at time $-T$. The reflected workloads at zero increase with T and are dominated by the stationary homogeneous M/D/1 workload driven by all proposal points. The latter queue is stable and has finite moments of every order. The monotone limit therefore exists, is stationary by shift-equivariance, and has the required finite moments.

At a deterministic stationary time write V for workload. Integrate the pathwise identity

$$V(t)^2 - V(0)^2 = -2 \int_0^t V(s) ds + \int_{(0,t]} (2V(s-) + 1) N(ds).$$

Domination and bounded intensity make these terms integrable. The predictable compensator gives

$$w(c) = \epsilon/2 + \epsilon \mathbb{E}[X(0)V].$$

No independence between workload and environment and no PASTA assertion are used.

On the same arrivals define the infinite-server residual work

$$Y = \sum_{T_i \in (-1,0]} (1 + T_i).$$

Pathwise $Y \leq V$: a job of age $u < 1$ cannot have received more than u units of service. Conditional Poisson expectation gives $\mathbb{E}Y = \epsilon/2$ and $\mathbb{E}[X(0)Y] = \epsilon K(c)$. Scalar stationary autocorrelation is even, so the backwards time argument in this calculation equals the displayed $K(c)$.

Let $D = V - Y \geq 0$ and $d = \mathbb{E}D = w(c) - \epsilon/2$. Then exactly

$$d = \epsilon^2 K(c) + \epsilon \mathbb{E}[X(0)D].$$

The pointwise bounds $0 \leq X \leq M$ imply

$$\epsilon^2 K(c) \leq d \leq \epsilon^2 K(c) + \epsilon M d,$$

which proves (1). Thus (1) controls the entire remainder, not merely one term in an asymptotic expansion. In particular,

$$w(c_2) - w(c_1) \geq \epsilon^2 \left[K(c_2) - \frac{K(c_1)}{1 - \epsilon M} \right]. \quad (2)$$

3 Full-support 64-state Markov generator

Let $n = 64$, $\theta = 2\pi/n$, $r = 1/\sin\theta$, and $\delta = 1/2000$. Let S be the cyclic permutation matrix satisfying $(Sv)_j = v_{j+1 \bmod n}$, and let Π have every entry equal to $1/n$. Set

$$Q = r(S - I) + \delta(\Pi - I). \quad (3)$$

One interpretation is forward moves around a cycle at rate r , with independent uniform redraws of the state at rate δ . A redraw that chooses the same state is a null event. Actual off-diagonal rates are

$$q_{j,k} = r \mathbf{1}_{\{k=j+1 \bmod n\}} + \frac{1}{128000}, \quad k \neq j.$$

The diagonal is minus the off-diagonal row sum. Every off-diagonal entry is strictly positive. Row and column sums vanish, so the stationary distribution is uniform. The finite chain is irreducible and mixing.

Set

$$x_j = 1 + \frac{1}{2} \cos \left(\frac{2\pi(j+1/4)}{64} \right), \quad X(t) = x_{J(t)}. \quad (4)$$

All 64 x_j are distinct: cosine equality would require either identical indices modulo n or $j+k+1/2$ to be a multiple of n , the latter being impossible. Therefore the scalar intensity observation $X(t)$ is itself a CTMC, not merely a non-Markov function of a hidden chain. Its mean is 1 and $1/2 < x_j < 3/2$.

Use $\epsilon = 1/2000$, $\lambda_c(t) = \epsilon x_{J(ct)}$, and unit services. The actual environmental generator is cQ and peak arrival intensity is below $3/4000$, so all compared queues are dominated by a stable homogeneous M/D/1 queue.

It lies outside stochastic-monotonicity assumptions

In the ordering of numerical intensities, $x_{62} < x_1 < x_{63} < x_0$. For the upper set $H = \{x_{63}, x_0\}$, both states x_{62} and x_1 start below H . From state 62 the forward move enters H , but from state 1 it does not. The uniform-reset contribution to the instantaneous probability of entering H is identical from these two states. Hence

$$P_{62}(X(t) \in H) - P_1(X(t) \in H) = rt + O(t^2) > 0$$

for sufficiently small t . Thus the scalar chain is not stochastically monotone; this is not a counterexample to the established positive theorem under double stochastic monotonicity.

4 Covariance and exact sign certificate

On the first Fourier mode, Π acts as zero and S acts by $e^{i\theta}$. Its generator eigenvalue is

$$r(e^{i\theta} - 1) - \delta = -\alpha + i, \quad \alpha = \tan(\pi/64) + \delta.$$

Averaging over uniform initial states yields

$$\mathbb{E}[X(0)X(t)] = 1 + \frac{1}{8}e^{-\alpha t} \cos t, \quad t \geq 0.$$

Consequently

$$K(c) = \frac{1}{2} + \frac{1}{8}I_\alpha(c), \quad I_\alpha(c) = \operatorname{Re} \frac{e^z - 1 - z}{z^2}, \quad z = (-\alpha + i)c. \quad (5)$$

For completeness, the following certificate uses only elementary inequalities and rational arithmetic, not a floating-point sign test.

Using $3 < \pi < 22/7$, $\sin x < x$ and $\cos x > 1 - x^2/2$, with $u = 11/224$,

$$0 < \alpha < \frac{u}{1 - u^2/2} + \frac{1}{2000} = \frac{9956231}{200462000} < \frac{1}{20}.$$

Define $A = \alpha/(1 + \alpha^2)$ and $C = (1 - \alpha^2)/(1 + \alpha^2)^2$. Then $A < \alpha$ and $99/100 < C < 1$. At multiples of π ,

$$I_\alpha(k\pi) = \frac{A}{k\pi} + C \frac{1 - (-1)^k e^{-\alpha k\pi}}{(k\pi)^2}.$$

Using $e^{-x} \geq 1 - x$ and $1 - e^{-x} \leq x$,

$$\begin{aligned} I_\alpha(3\pi) - I_\alpha(2\pi) &> -\frac{1}{360} + \frac{99}{100} \left[\frac{2}{9(22/7)^2} - \frac{1}{72} \right] \\ &= \frac{91}{15840} > \frac{1}{200}. \end{aligned}$$

The bracket is positive, which justifies multiplying its lower bound by the lower bound on C . Therefore

$$K(3\pi) - K(2\pi) > \frac{1}{1600}.$$

Also $I_\alpha(2\pi) \leq \alpha/\pi < 1/60$, giving $K(2\pi) < 241/480$. Substitution into (2) gives

$$\begin{aligned} w(3\pi) - w(2\pi) &> \frac{1}{2000^2} \left[\frac{1}{1600} - \frac{(3/4000)(241/480)}{1 - 3/4000} \right] \\ &= \frac{1587}{25\,580\,800\,000\,000} > 0. \end{aligned}$$

For orientation, direct formula evaluations give $K(2\pi)$ approximately 0.50182682190600814 and $K(3\pi)$ approximately 0.50292850997295202. These decimals are not the sign certificate.

Robustness

The certificate is strict. In a neighborhood of this full-support generator the stationary distribution, finite-time matrix exponentials and K -values depend continuously on the parameters. Intensity labels remain positive and distinct, and the subcritical peak bound remains valid. Normalize perturbed intensity vectors by their stationary mean if keeping $\mathbb{E}X = 1$ fixed. The strict separation in (2) therefore persists in a sufficiently small open neighborhood. No size of that neighborhood is asserted here.

The deterministic service assumption is allowed by the stated problem. This note does not establish a reversal for exponential service times.

5 A smaller all-rational 24-state example

Let $Q = S - I$ for a directed 24-cycle with unit forward rates, and use uniform stationarity. Define x by dividing the following exact integers by 1000:

$$\begin{aligned} &1499, 1473, 1416, 1330, 1221, 1098, 967, 839, \\ &722, 624, 552, 510, 501, 527, 584, 670, \\ &779, 902, 1033, 1161, 1278, 1376, 1448, 1490. \end{aligned}$$

These values define the model; they are not uncertain decimal approximations. They are distinct, positive, average exactly 1 and are below $3/2$. Thus $x_{J(t)}$ itself is an irreducible mixing CTMC. Take $\epsilon = 1/1000000$, unit services, and compare the actual environmental generators $29Q$ and $30Q$.

The verifier obtains rational enclosures of

$$K(c) = \frac{1}{24} x^T \left[\sum_{m=0}^{\infty} \frac{c^m Q^m}{(m+2)!} \right] x. \quad (6)$$

This follows by termwise integration of the uniformly convergent matrix exponential on $[0, 1]$. With terms through $m = L$ and $\|Q\|_\infty = 2$, the error in K is at most

$$\frac{M(2c)^{L+1}}{(L+3)!} \frac{1}{1-2c/(L+4)} \quad (7)$$

when $2c < L + 4$. Here $\|(x/24)^T\|_1 = 1$ and $\|x\|_\infty \leq M$, which give the scalar norm bound. The ratio of successive omitted terms is at most $2c/(L + 4)$, which gives the geometric tail bound.

With $L = 240$, exact `Fraction` arithmetic proves the outward-rounded bounds

$$K(29) < \frac{503796}{1000000}, \quad K(30) > \frac{503798}{1000000}.$$

The omitted-series widths are less than 7×10^{-51} and 3×10^{-47} respectively, but decimal error estimates are not used for the certificate. Applying (2),

$$w(30) - w(29) > \frac{1244303}{9999985000000000000000000} > \frac{1}{10^{18}} > 0.$$

This is a much smaller absolute certified effect than in the 64-state example. The gain here is fewer states and entirely rational input data, not practical performance magnitude. The smaller state count was found by searching a family of cycle-based examples. Neither failure to find smaller examples nor numerical exploration establishes global minimality.

6 Reproduction and status

From the repository root, run

```
python3 verify/run_all.py
```

The scripts use Python 3.10 or newer and its standard library. The runner checks the periodic certificate in Appendix A, the 64-state elementary rational certificate, and the 24-state rational matrix-series enclosures. Decimal displays are diagnostics; all comparisons establishing the certificates use exact arithmetic. The final output on success is **ALL CHECKS PASSED**.

The supplied audit also reports numerical cross-checks of the 64-state generator and its semi-group using NumPy/SciPy, a closed-form integral check with mpmath at 70 decimal digits, and direct matrix-exponential quadrature for the 24-state generator. Those optional numerical routines are not included in this repository. Such numerical checks are separate checks of calculations, not independent review of the probability proof.

7 Research context and limitations

Leskelä (2022), Problem 1, asks for universal modulation-speed workload monotonicity for stationary ergodic intensities and for a characterization of when it holds. Its discussion records positive results for special finite-state CTMCs, including those whose generator and time reversal are both stochastically monotone. These examples address the universal assertion even within finite-state CTMC environments; they do not contradict those restricted positive theorems.

The simplified random-phase sinusoid is non-Markov and nonmixing. The cycle construction in `docs/unrestricted_modulation_speed_v2.md` already avoids those limitations. The full-support refinement above adds positive rates between every pair of distinct states; the 24-state construction reduces state count with an exact computational certificate.

The first Ross conjecture, concerning comparison with a constant-rate Poisson queue, is a different assertion. A nonmonotone dependence on speed does not contradict that lower-bound theorem.

A targeted search and inspection of Leskelä's precise statement do not establish originality. Earlier periodic-input results and earlier Markov-modulated queue results remain relevant prior art. Even an old periodic counterexample can be important prior art for a Markov approximation construction. No claim of first discovery, global minimality, full characterization, or all-load reversal is made. Novelty relative to Rolski (1987, 1989), Lemoine (1989), Heyman (1982), and Miyoshi–Rolski (2004) is not established.

A Random-phase periodic certificate

The following specialization is recorded in `docs/upper_bound_audit.md`. Let

$$X(t) = 1 + \cos(t + \Theta), \quad \Theta \sim \text{Uniform}[0, 2\pi), \quad M = 2.$$

Uniform phase gives stationarity. The continuous-time shifts act as every rotation of the phase circle, so any event invariant under all shifts has probability zero or one: the intensity is ergodic. It is not mixing. The service law is degenerate iid and independent of arrivals.

Direct integration gives

$$\mathbb{E}[X(0)X(t)] = 1 + \frac{1}{2} \cos t, \quad K(c) = \frac{1}{2} + \frac{1 - \cos c}{2c^2}.$$

At $c_1 = 2\pi$ and $c_2 = 3\pi$,

$$K(c_1) = \frac{1}{2}, \quad K(c_2) = \frac{1}{2} + \frac{1}{9\pi^2} > \frac{1}{2} + \frac{1}{90}.$$

The strict elementary estimate uses $\pi^2 < 10$, for example from $\pi < 22/7$. Writing ρ for the mean arrival rate, (1) yields

$$\begin{aligned} w_\rho(3\pi) - w_\rho(2\pi) &> \rho^2 \left(\frac{1}{90} - \frac{\rho}{1 - 2\rho} \right) \\ &= \frac{\rho^2(1 - 92\rho)}{90(1 - 2\rho)}. \end{aligned}$$

In particular, every $0 < \rho < 1/92$ is certified by this coarse bound. This is a sufficient range, not a necessary range for the reversal itself.

At $\rho = 1/100$,

$$w_{1/100}(2\pi) \leq \frac{99}{19600}, \quad w_{1/100}(3\pi) > \frac{2273}{450000},$$

so

$$w_{1/100}(3\pi) - w_{1/100}(2\pi) > \frac{1}{11025000} > 0.$$

At $\rho = 1/1000$,

$$w_{1/1000}(2\pi) \leq \frac{999}{1996000}, \quad w_{1/1000}(3\pi) > \frac{22523}{45000000},$$

so

$$w_{1/1000}(3\pi) - w_{1/1000}(2\pi) > \frac{227}{22455000000} > 0.$$

Acknowledgment of AI assistance

AI tools assisted with the mathematical derivation, writing, and verification code; the author is responsible for the content.

References

- [1] L. Leskelä (2022). *Ross’s second conjecture and supermodular stochastic ordering*. Queueing Systems 100, 213–215. doi:10.1007/s11134-022-09824-0; arXiv:2204.11856. The supplied audit reports direct inspection of the problem statement and CTMC discussion.
- [2] N. Baeuerle and T. Rolski (1998). *A monotonicity result for the workload in Markov-modulated queues*. Journal of Applied Probability 35(3), 741–747. doi:10.1239/jap/1032265221. The positive theorem’s hypothesis is discussed in reference [1]; this audit does not claim a fresh full-text review of reference [2].
- [3] N. Miyoshi and T. Rolski (2004). *Ross-type conjectures on monotonicity of queues*. Australian & New Zealand Journal of Statistics 46(1), 121–131. Contains positive comparisons under regularity assumptions; not evidence that every finite-state Markov environment has the ordering.